Course Structure

Course Code	BCA20110				
Course Category	Program Major				
Course Title	Programming in JAVA				
Teaching Scheme and Credits Weekly Load hrs	L	T	Laborat ory/Practical	Project	Total
	3	-	2	-	5
Credits	3	-	1	-	4
Assessment Schema Code	TL3				
Pre-requisites: Need basics of Programming and Object-Oriented Programming concepts.					

Course Objectives

Knowledge: To understand Java programming concepts with respect to

i) Object Oriented Programming

ii) ii) Data Structures in Java Programming

Skills: Design skills of

i) Desktop Applications Development

ii) ii) Object oriented programming

Attitude: To develop following

i) Programming Skills

ii) Build Stand-alone Application in Complex situation

Course Outcomes:

a) Identify classes, objects, class members and relationships for a given problem.
b) Design end to end applications using object-oriented constructs.
c) Apply collection classes for storing java objects.
d) Use Java APIs for program development.
e) Handle abnormal termination of a program using exception handling.

Course Contents

Unit 1: Introduction to Java: 4

A Short History of Java, Features of Java, Java Environment – Compiler, Interpreter, JVM, Simple java, program, Types of Comments, Declaring single and multi-dimensional arrays, Accepting input using Command line arguments, Accepting input from console (Using BufferedReader and Scanner class)

Unit 2: Classes and Objects:8

Defining Your Own Classes, Access Specifiers (public, protected, private, default), Array of Objects, Constructor, Overloading Constructors and use of, this" Keyword static blocks, static Fields and static methods, Predefined classes - Object class(toString(), hashcode()), Garbage Collection (finalize() Method)
Packages-Creating, Accessing and using Packages

Unit 3: Inheritance:10

Inheritance Basics (extends Keyword) and Types of Inheritance, Superclass, Subclass and use of Super Keyword, Method Overriding and runtime polymorphism, Use of final keyword related to variable, method and class, Use of abstract class and abstract methods

Interface and Packages-Defining and Implementing Interfaces, Runtime polymorphism using interface

Unit 4: Collections:8

Wrapper Classes, Introduction to the Collection framework, List – ArrayList, LinkedList and Vector, Set - HashSet, TreeSet, and LinkedHashSet, Map – HashMap, LinkedHashMap, Hashtable and TreeMap Interfaces such as Iterators, ListIterators, Enumerations

Unit 5:Exception Handling : 4

Exception class, Checked and Unchecked exception, Catching exception and exception handling – try, catch, finally, throw and throws, multiple catch block, Creating user defined exception

Unit 6: I/O: 3

String class(basic methods) , String Buffer class DataOutputStream class

Unit 7: Swing – 8

What is Swing? The MVC Architecture and Swing, Layout Manager and Layouts, The JComponent class Components – JLabel, JButton, JText, JTextArea, JCheckBox, JRadioButton, JUst, JComboBox, JMenu and JPopupMenu Class, JMenuItem, Dialogs (Message, confirmation, input), JFileChooser, Event Handling: Event sources, Listeners – ActionListener, ItemListener, Mouse and Keyboard Event Handling, Adapters – MouseAdapter, Key Adapter, Anonymous inner class

Learning Resources:
Reference Books:

1. Java 2 programming black books, Steven Horlzner

Supplementary Reading:

1. Complete reference Java by Herbert Schildt(5th edition)
2. Java Complete Reference Patric Naughton, Herbert Schildt, TMH,7th Ed.
3. Complete Reference- J2EE Jim Keogh, TMH.
4. Inside Servlets- Dustine R Callway
5. Programming with Java, A primer, Fourth edition, By E. Balagurusamy
6. Core Java Volume-1-Fundamentals, Eighth Edition, Cay S. Horstmann, Gary Cornell,

Additional Reading

Websites :

https://www.tutorialspoint.com

https://www.javatpoint.com

MOOCS : Spoken Tutorial, Khan Academy

Pedagogy:

- Practical learning
- project handling
- problem solving assignment
- conceptual and contextual learning
- PPT Presentation & Interactive discussion

Laboratory Experiments

- Basic Java concepts, constants, operators, decisions and loops
- Classes, Objects, Constructors, Packages
- Inheritance, Aggregation, Interface, Polymorphism
- Collection Framework
- Exception Handling
- I/O Handling
- GUI Development using Swing